

SAMPLE QUESTIONS:

Q: Set the owner of `/root/203508.txt` to the `operator` user without changing anything else.

A: `chown operator /root/203508.txt`

`chown` changes only the owner. Omitting the group (no `:group`) ensures nothing else is modified.

Q: The file `/root/203510` contains a LUKS-encrypted filesystem with password `cbzmy`. Copy its contents to `/root`.

A: `cryptsetup open /root/203510 myluks --type luks`
enter password: `cbzmy`

```
mount /dev/mapper/myluks /mnt
cp -r /mnt/* /root/
umount /mnt
cryptsetup close myluks
```

Opens the LUKS container, mounts it, copies all contents to `/root`, then cleanly closes it.

Q: Create a self-signed certificate for CN `self.my-factory.eu` and save it to `/root/self.my-factory.eu.crt`.

A: `openssl req -x509 -newkey rsa:2048 -nodes -keyout /root/self.my-factory.eu.key \`
`-out /root/self.my-factory.eu.crt -days 365 -subj "/CN=self.my-factory.eu"`

Generates a self-signed certificate and key in one command; `-subj` sets the CN directly without an interactive prompt.

Q4: Create user **user203511** and lock their password.

```
useradd user203511  
passwd -l user203511
```

useradd creates the user; **passwd -l** locks the password by prefixing **!** in **/etc/shadow**.

Q5: Ensure no user has access to **/root/203512.key** by removing all privilege-related (ACL) entries.

```
chmod 000 /root/203512.key  
setfacl -b /root/203512.key
```

chmod 000 removes all standard permissions; **setfacl -b** strips all ACL entries from the file.

Q6: Find the directory inside **/root/203513/** not owned by **root**, write its full path to **/root/203513.ans**.

```
find /root/203513/ -type d ! -user root > /root/203513.ans
```

find searches for directories (**-type d**) whose owner is not root, saving the absolute path to the answer file.

Q7: Allow TCP port 9000 permanently in the **dmz** zone using firewallld.

```
firewall-cmd --zone=dmz --add-port=9000/tcp --permanent  
firewall-cmd --reload
```

--permanent persists the rule across reboots; **--reload** applies it immediately.

Q8: Allow the owner of the given private key to log in as **root** via SSH.

```
# Save the private key locally, then extract its public key:
ssh-keygen -y -f private.key >> /root/.ssh/authorized_keys
chmod 600 /root/.ssh/authorized_keys
```

ssh-keygen -y derives the public key from the private key; appending it to **authorized_keys** grants root SSH access to whoever holds that private key.

Q9: Find which file was signed by **/root/203514.sign** using SHA256 and the given public key, then sign the same file with the given private key and save to **/root/203514.ans**.

```
# Save the public key to a file
cat > /tmp/pub.key << 'EOF'
-----BEGIN PUBLIC KEY-----
MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBABJWEsdK3icjH7ZpEJDHwo8wm4l0KPu
5
ITIBwQAf5bqzFX6mZrhhgAvL0Jf+GCwgpr7YUqfoDt9hzZVwZ+7W8tUCAwEAAQ==
-----END PUBLIC KEY-----
EOF
```

```
# Verify each file to find which one matches the signature
for f in /root/203514A.txt /root/203514B.txt /root/203514C.txt; do
    openssl dgst -sha256 -verify /tmp/pub.key -signature /root/203514.sign "$f" && echo
    "MATCH: $f"
done
```

```
# Save the private key, then sign the matched file (replace 203514X.txt with the match)
cat > /tmp/priv.key << 'EOF'
<paste private key here>
EOF
```

```
openssl dgst -sha256 -sign /tmp/priv.key -out /root/203514.ans /root/203514X.txt
```

openssl dgst -verify checks each file against the signature; once matched, **-sign** with the private key generates the new signature.

Q10: Set **/root/203515.txt** to read-only.

```
chmod 444 /root/203515.txt
```

Sets read permission for owner, group, and others; removes all write/execute bits.

Q11: Create **/dev/sdc1** on **/dev/sdc**, format with ext2, mount read-only to **/mnt**.

```
fdisk /dev/sdc # press: n, p, 1, Enter, Enter, w
mkfs.ext2 /dev/sdc1
mount -o ro /dev/sdc1 /mnt
```

fdisk creates the partition, **mkfs.ext2** formats it, **-o ro** mounts it read-only.

Q12: Decrypt **/root/203516.enc** using password **203516**, **des-ede**, and **pbkdf2**.

```
openssl enc -des-ede -d -pbkdf2 -in /root/203516.enc -out /root/203516.txt -pass
pass:203516
```

-d decrypts using the specified algorithm and key derivation function.

Decrypt the **/root/17218.enc** file by using **17218** as the key and **0** as the initialization vector. The file was encrypted by using the aes-128-cbc algorithm. Save the result as **/root/17218.txt**.

```
openssl aes-128-cbc-d-in /root/17218.enc-K 17218-iv 0-out /root/17218.txt
xxd /root/17218.enc
```

Decrypt the **/root/17205.enc** file by using the **17205** password and the des-ede algorithm. Save the result as **/root/17205.txt**.

```
openssl des-ede-d-in /root/17205.enc-k 17205-out /root/17205.txt
xxd /root/17205.txt
```

Q13: Give **operator** user read-only ACL on **/root/203517.txt**.

```
setfacl -m u:operator:r /root/203517.txt
```

-m u:operator:r adds only a read ACL entry without touching existing permissions.

Q14: Set account expiry for `user203518` to `2025-01-01`.

`chage -E 2025-01-01 user203518`

`chage -E` sets the account expiration date directly.

PAST QUESTIONS:

Sign the certificate signature request stored in `/root/my-factory.csr` using the certificate stored in `/root/my-root-ca.crt` and the corresponding private key stored in `/root/my-root-ca.key`. Save the signed certificate to the file: `/root/my-factory.crt`.

`openssl x509-req-in /root/my-factory.csr-CA /root/my-root-ca.crt-CAkey /root/my-root-ca.key-CAcreateserial-out /root/my-factory.crt #-days 365`

unlock an account

`passwd-u user265273`

Find the folder inside the `/root/17107/` folder, which has an enabled sticky bit. Save the full path of this folder into the `/root/17107.ans` file.

`find /root/17115-perm-1000 >/root/17115.ans`

Create a file of size 99999999 bits

`dd if=/dev/zero of=/file/path bs=99999999 count=1`

```
# setuid=4000 setgid=2000 sticky=1000
chmod 4755 file # rwsr-xr-x (setuid)
chmod 2755 file # rwxr-sr-x (setgid)
chmod 1777 dir # rwxrwxrwt (sticky)
chmod 3460 file # r--rws--T
```

Lowercase `s/t` = special bit + execute set

Uppercase `S/T` = special bit set, execute NOT set